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(54) **BIOLOGICAL MATERIAL EXTRACTION
MAGNETIC BEAD**

(52) **U.S. CL.**
CPC **G01N 1/34** (2013.01)

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(57) **ABSTRACT**

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A biological material extraction magnetic bead includes: a magnetic metal particle; an inorganic oxide layer that covers a surface of the magnetic metal particle and contains an inorganic oxide; a base layer that is provided on a side opposite to the magnetic metal particle of the inorganic oxide layer and contains a metal element forming a gold ion reducing agent or a catalyst for a gold ion reduction reaction; a gold layer that covers a surface of the base layer and contains gold; and an immobilization layer that is bound to the gold layer via an Au—S bond and contains a ligand or a ligand-reactive group. A mass ratio M/Au of a content of the metal element to a content of the gold is 0.03 or more and 19.0 or less.

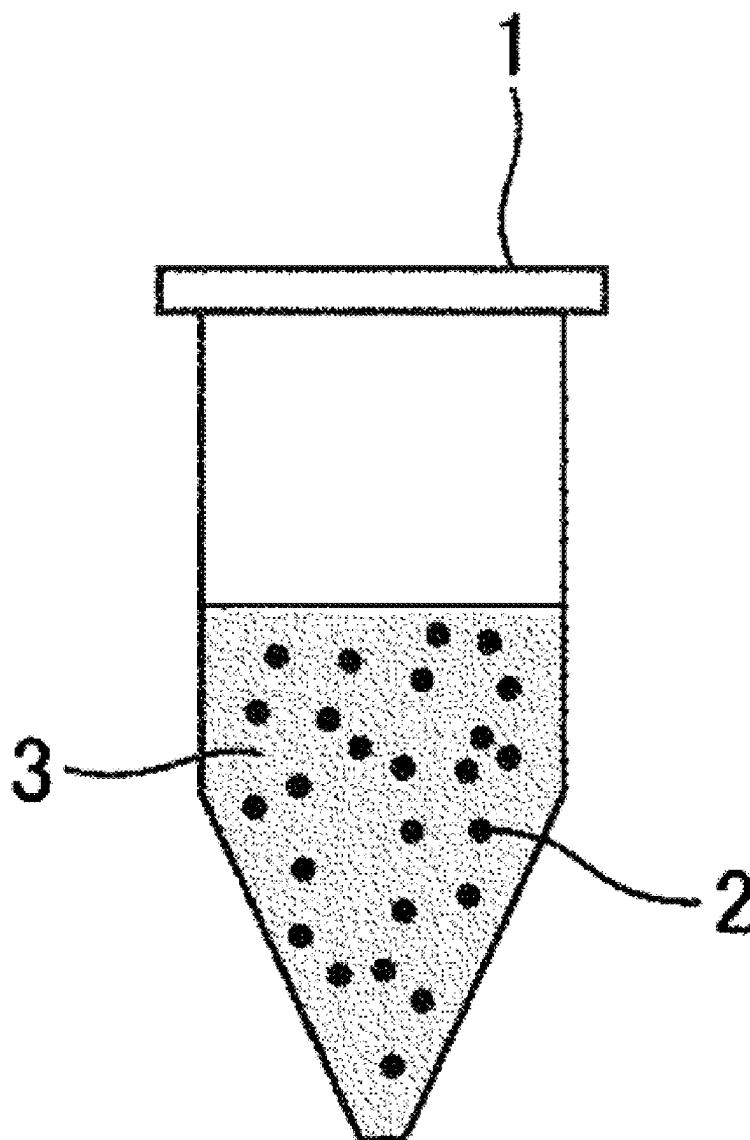


FIG. 1

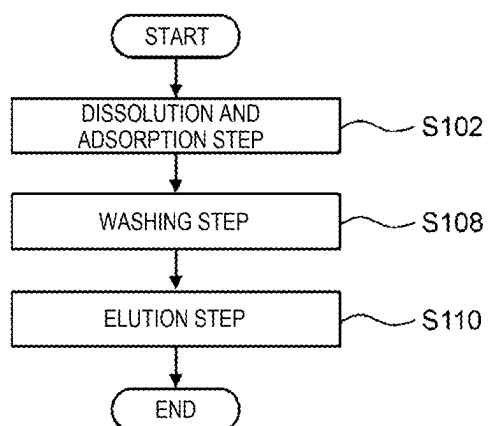


FIG. 2

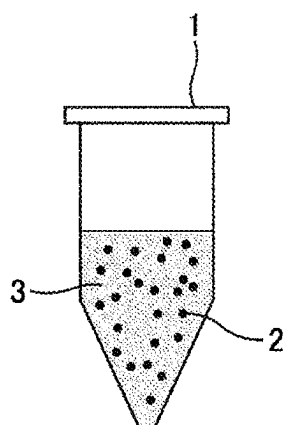


FIG. 3

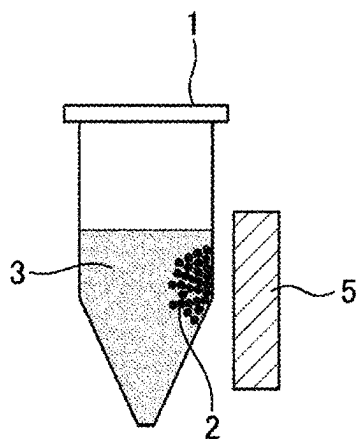


FIG. 4

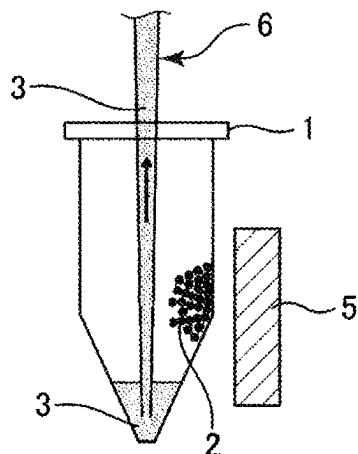


FIG. 5

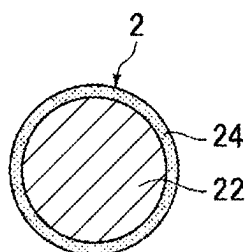


FIG. 6

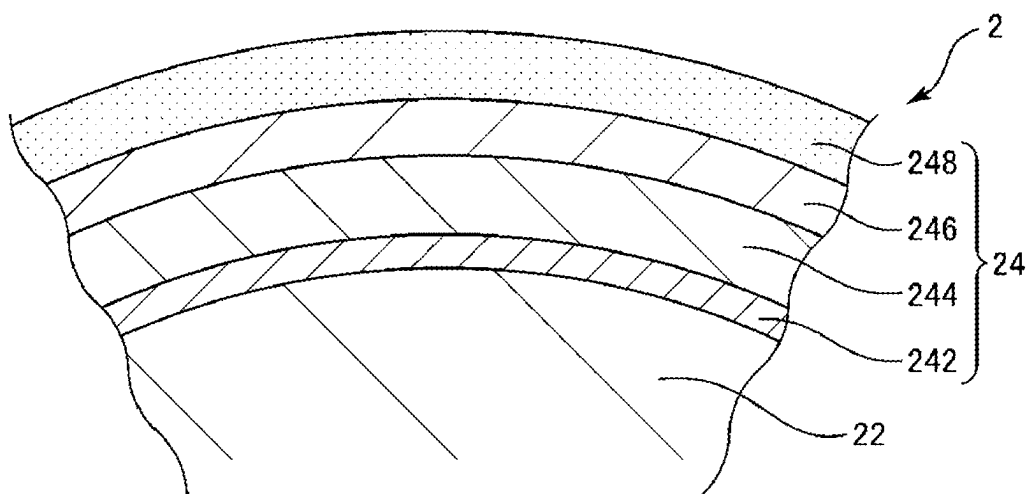


FIG. 7

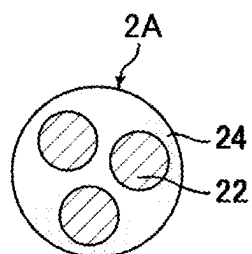
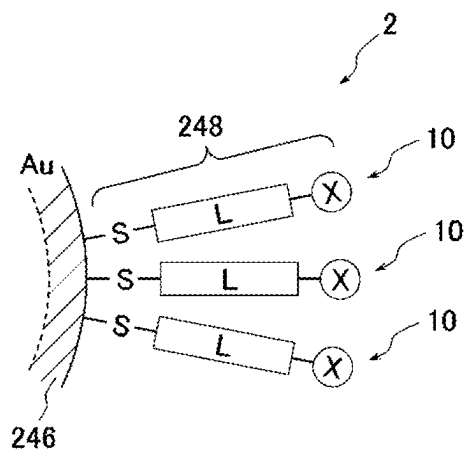


FIG. 8



BIOLOGICAL MATERIAL EXTRACTION MAGNETIC BEAD

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-032914, filed Mar. 5, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a biological material extraction magnetic bead.

2. Related Art

[0003] In recent years, in the fields of diagnosis and biological science in the medical field, there is an increasing demand for examination of a biological material. Among biological material examination techniques, a polymerase chain reaction (PCR) method is a method of extracting a nucleic acid such as deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) and specifically amplifying and detecting the nucleic acid. In such a process of examining the biological material, it is first necessary to extract a substance to be examined from a specimen. In the extraction of the biological material, for example, an extraction carrier having a binding ability for the biological material is used.

[0004] JP-A-2017-122728 discloses a ligand as a binding species having a binding ability to a cell surface receptor or a virus. It is also disclosed that a biomolecule is detected using a colloid holding the ligand. It is also disclosed that a self-assembled monolayer containing thiol is formed at a surface of a particle covered with gold, and the self-assembled monolayer presents a binding partner of the ligand. Further, it is disclosed that a magnetic bead is used as the particle.

[0005] In the extraction of the biological material, various physical and chemical loads are applied to the extraction carrier. When such loads are applied to the colloid holding the ligand disclosed in JP-A-2017-122728, the biological material bound to the extraction carrier may be detached or the extraction carrier may aggregate, and thus biological material extraction efficiency may decrease.

[0006] The colloid disclosed in JP-A-2017-122728 has room for improvement in dispersibility in a dispersion medium. When the dispersibility is low, efficiency of capturing the biological material decreases, and as a result, the biological material extraction efficiency decreases.

[0007] Further, when the magnetic bead is used as the particle disclosed in JP-A-2017-122728, metal ions contained in the magnetic bead may elute into the dispersion medium and inhibit examination of the biological material.

SUMMARY

[0008] Therefore, an object is to provide a biological material extraction magnetic bead that has favorable biological material extraction efficiency and is less likely to lower examination accuracy of an extracted biological material.

[0009] A biological material extraction magnetic bead according to an application example of the disclosure includes:

[0010] a magnetic metal particle;

[0011] an inorganic oxide layer that covers a surface of the magnetic metal particle and contains an inorganic oxide;

[0012] a base layer that is provided on a side opposite to the magnetic metal particle of the inorganic oxide layer and contains a metal element forming a gold ion reducing agent or a catalyst for a gold ion reduction reaction;

[0013] a gold layer that covers a surface of the base layer and contains gold; and

[0014] an immobilization layer that is bound to the gold layer via an Au—S bond and contains a ligand or a ligand-reactive group, in which

[0015] a mass ratio M/Au of a content of the metal element to a content of the gold is 0.03 or more and 19.0 or less.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a process flowchart showing an example of a biological material extraction method.

[0017] FIG. 2 is a schematic diagram showing the biological material extraction method shown in FIG. 1.

[0018] FIG. 3 is a schematic diagram showing the biological material extraction method shown in FIG. 1.

[0019] FIG. 4 is a schematic diagram showing the biological material extraction method shown in FIG. 1.

[0020] FIG. 5 is a cross-sectional view showing a magnetic bead according to an embodiment.

[0021] FIG. 6 is a partially enlarged view of a coating film shown in FIG. 5.

[0022] FIG. 7 is a cross-sectional view showing a modification of the magnetic bead in FIG. 5.

[0023] FIG. 8 is a schematic diagram showing a state in which a compound having a functional site is bound to a gold layer via an Au—S bond.

DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, a preferred embodiment of biological material extraction magnetic bead according to the disclosure will be described in detail with reference to the accompanying drawings.

[0025] The biological material extraction magnetic bead according to the embodiment is a particle group that adsorbs a biological material and is used for magnetic separation. Magnetic separation is a technique of separating a solid phase and a liquid phase by applying an external magnetic field to a container in which a solid phase containing a biological material extraction magnetic bead and a liquid phase containing a dispersion medium are charged to magnetically attract the solid phase. In the following description, the biological material extraction magnetic bead may be simply referred to as a “magnetic bead”.

[0026] The biological material refers to substances such as nucleic acids such as DNA or RNA, proteins, saccharides, various cells such as cancer cells, peptides, bacteria, and viruses. The nucleic acid may be present in a state of being contained in a biological sample such as a cell or biological tissue, a virus, or a bacterium. A biological material extraction method is a method of extracting such a biological

material through each step of, for example, dissolution and adsorption, washing, and elution, and in such a process, the biological material can be purified and extracted using the above-described magnetic separation.

1. Biological Material Extraction Method

[0027] Hereinafter, an example of the biological material extraction method using magnetic separation will be described. In the following description, a case where the biological material is a nucleic acid will be described as an example.

[0028] FIG. 1 is a process flowchart showing an example of the biological material extraction method. FIGS. 2 to 4 are schematic diagrams showing the biological material extraction method shown in FIG. 1.

[0029] The biological material extraction method shown in FIG. 1 includes a dissolution and adsorption step S102, a washing step S108, and an elution step S110. Hereinafter, each step will be sequentially described.

1.1. Dissolution and Adsorption Step

[0030] In the dissolution and adsorption step S102, a liquid 3, which contains a dissolution and adsorption liquid and a specimen sample that contains a nucleic acid, and a magnetic bead 2 are charged into a container 1 shown in FIG. 2. Then, contents in the container 1 are mixed. Accordingly, the magnetic bead 2 is dispersed in the liquid 3 in the container 1 as shown in FIG. 2. The nucleic acid is usually enclosed in a cell membrane or a nucleus. Therefore, the nucleic acid is extracted from the specimen sample when a so-called outer shell of the cell membrane or the nucleus is dissolved and removed by a dissolving action of the dissolution and adsorption liquid. Thereafter, the nucleic acid is adsorbed to the magnetic bead 2 by an adsorption action of the dissolution and adsorption liquid. The magnetic bead 2 where the nucleic acid is adsorbed is separated from a liquid phase through a magnetic separation operation and a liquid discharge operation to be described later.

[0031] As the dissolution and adsorption liquid, for example, a liquid containing a chaotropic substance is used. The chaotropic substance has a function of generating chaotropic ions in an aqueous solution, reducing an interaction between water molecules, and thereby destabilizing a structure, and thus contributes to adsorption of the nucleic acid to the magnetic bead 2.

[0032] Particularly, when RNA among nucleic acids is extracted, it is preferable to acidify the liquid 3 in the container 1 by adding an acid or the like. An RNA monomer contains ribose and thus is more soluble in a polar solvent than DNA. Therefore, the acid is added to the liquid 3 containing the specimen sample from which the outer shell is dissolved and removed to acidify the liquid 3, and then a nonpolar solvent such as phenol or chloroform is added. Accordingly, DNA migrates to the non-polar solvent whereas RNA remains in the polar solvent. As a result, RNA and DNA can be separated, and RNA can be extracted.

[0033] When RNA is extracted, a pH of the liquid 3 in the container 1 is preferably 5.0 or less, more preferably 2.0 or more and 4.0 or less. Accordingly, an ionization equilibrium of a phosphate group contained in RNA is biased to a hydroxy group, and thus RNA is particularly easily dissolved in the polar solvent. Therefore, extraction efficiency of RNA can be further improved.

1.1.1. Magnetic Separation Operation

[0034] In the dissolution and adsorption step S102, an external magnetic field acts on the magnetic bead 2 where the nucleic acid is adsorbed, and the magnetic bead 2 is magnetically attracted. Accordingly, as shown in FIG. 3, the magnetic bead 2 is moved to an inner wall of the container 1 and fixed. As a result, the magnetic bead 2 in a solid phase can be separated from the liquid 3 in a liquid phase. In the specification, an operation of fixing the magnetic bead 2 by applying such an external magnetic field is referred to as the "magnetic separation operation".

[0035] Before the magnetic separation operation, the contents in the container 1 are stirred as necessary. Accordingly, a probability that the nucleic acid is adsorbed by the magnetic bead 2 increases. In the stirring, for example, a vortex mixer, hand shaking, or pipetting is used.

[0036] In the application of the external magnetic field, for example, a magnet 5 disposed beside the container is used. The magnet 5 may be an electromagnet or a permanent magnet. When the external magnetic field acts on the magnetic bead 2, the magnetic bead 2 moves toward the magnet 5.

[0037] After the magnetic separation operation, an acceleration may be applied to the container as necessary. Accordingly, the liquid 3 adhering to the magnetic bead 2 can be shaken off, and the unseparated liquid 3 can be reduced. The acceleration may be a centrifugal acceleration. A centrifugal separator may be used to apply the centrifugal acceleration.

1.1.2. Liquid Discharge Operation

[0038] In the dissolution and adsorption step S102, in a state in which the magnetic bead 2 is fixed to the inner wall of the container 1, as shown in FIG. 4, the liquid 3 accumulated at a bottom of the container 1 is suctioned and discharged by, a pipette 6. In the specification, such an operation of discharging the liquid 3 is referred to as the "liquid discharge operation". By the liquid discharge operation, the magnetic bead 2 where the nucleic acid is adsorbed remains in the container 1.

1.2. Washing Step

[0039] In the washing step S108, the magnetic bead 2 where the nucleic acid is adsorbed is washed. Washing refers to an operation of removing a foreign substance by bringing the magnetic bead 2 where the nucleic acid is adsorbed into contact with a washing liquid and then separating again in order to remove the foreign substance adsorbed to the magnetic bead 2.

[0040] Specifically, after the washing liquid is charged into the container 1 in which the magnetic bead 2 where the nucleic acid is adsorbed is charged, the magnetic separation operation and the liquid discharge operation described above are performed again.

[0041] Among these, in the magnetic separation operation, first, the magnetic bead 2 and the washing liquid in the container 1 are stirred. Accordingly, the washing liquid comes into contact with the magnetic bead 2, and the magnetic bead 2 where the nucleic acid is adsorbed is washed. In the stirring, for example, a vortex mixer, hand shaking, or pipetting is used. At this time, the application of the external magnetic field may be temporarily turned off.

Accordingly, since the magnetic bead **2** is redispersed in the washing liquid, washing efficiency can be further improved.

[0042] Next, as the liquid discharge operation, the washing liquid accumulated at the bottom of the container **1** is discharged in a state in which the magnetic bead **2** is fixed to the inner wall of the container **1**. The magnetic bead **2** is washed by performing the above-described supply and discharge of the washing liquid at least once. Accordingly, the foreign substance can be removed with high accuracy.

[0043] The washing liquid is not particularly limited as long as the washing liquid is a liquid that does not promote elution of the nucleic acid and does not promote binding of the foreign substance to the magnetic bead **2**, and examples thereof include organic solvents such as ethanol, isopropyl alcohol, and acetone, aqueous solutions thereof, and low-salt-concentration aqueous solutions.

[0044] The washing liquid may contain a surfactant such as Triton (registered trademark), Tween (registered trademark), or SDS. The washing liquid may contain a chaotropic substance such as guanidine hydrochloride.

[0045] The washing step **S108** may be performed as necessary and may be omitted when washing is not necessary.

1.3. Elution Step

[0046] In the elution step **S110**, the nucleic acid adsorbed to the magnetic bead **2** is eluted into an elution liquid. The elution is an operation of transferring the nucleic acid to the elution liquid by bringing the magnetic bead **2** where the nucleic acid is adsorbed into contact with the elution liquid.

[0047] Specifically, after the elution liquid is charged into the container **1** in which the magnetic bead **2** where the nucleic acid is adsorbed is charged, the magnetic separation operation and the liquid discharge operation described above are performed again.

[0048] Among these, in the magnetic separation operation, first, the magnetic bead **2** and the elution liquid in the container **1** are stirred. Accordingly, the elution liquid comes into contact with the magnetic bead **2**, and the nucleic acid is eluted into the elution liquid. In the stirring, for example, a vortex mixer, hand shaking, or pipetting is used. At this time, the application of the external magnetic field may be temporarily turned off. Accordingly, since the magnetic bead **2** is redispersed in the elution liquid, elution efficiency can be further improved.

[0049] Next, as the liquid discharge operation, the elution liquid accumulated at the bottom of the container **1** is discharged in a state in which the magnetic bead **2** is fixed to the inner wall of the container **1**. Accordingly, the elution liquid containing the nucleic acid can be collected.

[0050] The elution liquid is not particularly limited as long as the elution liquid is a liquid that promotes the elution of the nucleic acid from the magnetic bead **2** where the nucleic acid is adsorbed, and for example, in addition to water such as sterilized water or pure water, a TE buffer, that is, an aqueous solution which contains 10 mM of Tris-HCl buffer and 1 mM of EDTA and which has a pH of about 8 is preferably used.

[0051] The elution liquid may contain a surfactant such as Triton (registered trademark), Tween (registered trademark), or SDS. Sodium azide may also be contained as a preservative.

[0052] In the elution step **S110**, the elution liquid may be heated. Accordingly, the elution of the nucleic acid can be

promoted. A heating temperature of the elution liquid is not particularly limited, and is preferably 70° C. or higher and 200° C. or lower, more preferably 80° C. or higher and 150° C. or lower, and still more preferably 95° C. or higher and 125° C. or lower.

2. Biological Material Extraction Magnetic Bead

[0053] Next, the magnetic bead **2** (the biological extraction magnetic bead to the material according embodiment) will be described. The magnetic bead **2** is a particle that has magnetism and whose surface exhibits a binding affinity with a biological material.

[0054] FIG. **5** is a cross-sectional view showing the magnetic bead **2** according to the embodiment. The magnetic bead **2** shown in FIG. **5** includes a magnetic metal particle **22** and a coating film **24**. A metal powder having magnetism is used as the magnetic metal particle **22**. At least a surface of the coating film **24** is formed of a substance or a chemical structure having a binding affinity with the biological material. In the specification, the magnetic bead **2** refers to a particle group or one particle constituting the particle group.

[0055] According to such the magnetic bead **2**, magnetic separation is available when the biological material is to be separated from a liquid phase in a state of being adsorbed. In the magnetic separation, as compared to another separation method, for example, centrifugal separation, a momentum applied to an adsorption body where the biological material is adsorbed is low. Therefore, a load applied to the adsorption body is reduced and separation of the biological material and a ligand can be prevented.

[0056] A saturation magnetization of the magnetic metal particle **22** is preferably 50 emu/g or more, more preferably 80 emu/g or more, and still more preferably 100 emu/g or more. The saturation magnetization is a magnetization value in a case where a magnetization exhibited by a magnetic material when a sufficiently large magnetic field is externally applied is constant regardless of the magnetic field. If the saturation magnetization is within the above range, a function as a magnetic material can be sufficiently exhibited. Specifically, since a movement speed of the magnetic bead **2** in a magnetic field can be increased, a time required for magnetic separation can be shortened. The saturation magnetization of the magnetic metal particle **22** affects an adsorption force when being fixed by an external magnetic field. If the saturation magnetization is within the above range, a sufficiently high adsorption force can be obtained, and therefore, when the liquid **3** is discharged in a state in which the magnetic bead **2** is fixed, the magnetic bead **2** can be prevented from being discharged together with the liquid **3**. Accordingly, it is possible to prevent a decrease in a nucleic acid yield due to a decrease in the magnetic bead **2**.

[0057] An upper limit value of the saturation magnetization of the magnetic metal particle **22** is not particularly limited, and is preferably 220 emu/g or less from the viewpoint of ease of selection of a material suitable for a balance between performance and cost.

[0058] The saturation magnetization of the magnetic metal particle **22** can be measured by a vibrating sample magnetometer (VSM) or the like. As the vibrating sample magnetometer, for example, TM-VSM1230-MHHL manufactured by Tamagawa Seisakusyo Co., Ltd. may be used. A maximum applied magnetic field when measuring the saturation magnetization is, for example, 0.5 T or more.

[0059] An average particle diameter D50 of the magnetic bead **2** is preferably 0.5 μm or more and 30 μm or less, more preferably 1 μm or more and 20 μm or less, still more preferably 2 μm or more and 15 μm or less, and particularly preferably 3 μm or more and 10 μm or less. If the average particle diameter D50 of the magnetic bead **2** is within the above-described range, a specific surface area of the magnetic bead **2** can be sufficiently large, and an attractive force and an adsorption force suitable for magnetic separation can be generated in the magnetic bead **2**. In addition, aggregation of the magnetic bead **2** can be prevented, and dispersibility can be improved. When the average particle diameter D50 of the magnetic bead **2** is less than the lower limit value, a magnetization value of the magnetic bead **2** is small, the magnetic bead **2** is likely to aggregate, and as a result, nucleic acid extraction efficiency may decrease. In addition, the movement speed of the magnetic bead **2** may decrease, and the time required for magnetic separation may increase. On the other hand, when the average particle diameter D50 of the magnetic bead **2** exceeds the upper limit value, since the specific surface area of the magnetic bead **2** is small, a sufficient amount of the nucleic acid cannot be adsorbed, and an extraction amount of the nucleic acid may decrease. In addition, the magnetic bead **2** is likely to settle, the magnetic bead **2** that can contribute to the extraction of the nucleic acid may decrease, and the nucleic acid extraction efficiency may decrease.

[0060] The average particle diameter D50 of the magnetic bead **2** can be obtained from a cumulative distribution curve obtained from a volume-based particle size distribution measured by a laser diffraction and dispersion method. Specifically, in the cumulative distribution curve, a particle diameter (median diameter) where a cumulative value is 50% from a small diameter side is the average particle diameter D50 of the magnetic bead **2**. Examples of an apparatus for measuring the particle size distribution by the laser diffraction and dispersion method include MT3300 series manufactured by MicrotracBEL. The method is not limited to the laser diffraction and dispersion method, and a method such as image analysis may be used.

[0061] A 90% particle diameter of the magnetic bead **2** is defined as D90. In the magnetic bead **2**, a ratio D90/D50 of the 90% particle diameter D90 to the average particle diameter D50 is preferably 3.00 or less, more preferably 2.00 or less, and still more preferably 1.75 or less. Accordingly, since a content of coarse particles is low, it is possible to prevent the coarse particles from attracting relatively small surrounding particles and aggregating to form an aggregate. When an aggregate is formed, the aggregate settles due to own weight, which may cause a decrease in the extraction efficiency and an increase in an examination time of the biological material accordingly. Therefore, when the ratio D90/D50 is within the above range, occurrence of such a problem can be prevented. When the ratio D90/D50 exceeds the upper limit value, the content of coarse particles is high, and therefore, even when the application of the external magnetic field is turned off, the dispersibility of the magnetic bead **2** decreases, and aggregation is likely to occur.

[0062] The 90% particle diameter D90 of the magnetic bead **2** can be obtained from a cumulative distribution curve obtained from a volume-based particle size distribution measured by a laser diffraction and dispersion method. Specifically, in the cumulative distribution curve, a particle

diameter where a cumulative value is 90% from a small diameter side is the 90% particle diameter D90 of the magnetic bead **2**.

[0063] A ratio $t/D50$ of t to D50, in which t is an average thickness of the coating film **24** and D50 is the average particle diameter of the magnetic bead **2**, is preferably 0.0001 or more and 0.05 or less, and more preferably 0.001 or more and 0.01 or less. When $t/D50$ is less than the lower limit value, a ratio of the thickness of the coating film **24** to a size of the magnetic metal particle **22** is excessively small, and thus the coating film **24** may be broken or peeled off when the magnetic bead **2** collides with each other or the magnetic bead **2** collides with the inner wall of the container **1** or the like. Therefore, an amount of the nucleic acid adsorbed and extracted by the surface of the coating film **24** decreases, and the extraction efficiency may decrease. When there are fragments of the peeled coating film **24** or the magnetic metal particle **22** in an extracted liquid, the fragments may be mixed as a foreign substance (contamination) at the same time when the nucleic acid is extracted. Further, the magnetic metal particle **22** may be exposed due to breaking and peeling of the coating film **24**, and elution of iron ions or the like may occur when the magnetic metal particle **22** comes into contact with an acidic solution or the like, resulting in a decrease in the nucleic acid extraction efficiency. On the other hand, when $t/D50$ exceeds the upper limit value, a volume ratio of the coating film **24** to an entire volume of the magnetic bead **2** is large, and a magnetization per unit volume of the magnetic bead **2** may decrease. Accordingly, the movement speed when the external magnetic field acts on the magnetic bead **2** decreases, and the time required for magnetic separation may increase.

[0064] The thickness of the coating film **24** can be measured from, for example, a cross-sectional observation image of the magnetic bead **2** observed with a transmission electron microscope or a scanning electron microscope. The average thickness t of the coating film **24** can be calculated by acquiring a plurality of observation images and averaging measured values from image processing or the like. For example, the average thickness t is a value obtained by measuring the thickness of the coating film **24** at five or more positions for one bead of the magnetic bead **2**, determining an average value thereof, and then averaging the average value for ten or more beads of the magnetic bead **2**. For example, intensities of a Si—K characteristic X-ray and an Fe—L characteristic X-ray may be compared using an analysis apparatus such as an energy dispersive X-ray spectroscopy (EDX), and the thickness of the coating film **24** may be calculated based on a comparison result. That is, as will be described later, when the coating film **24** contains silicon and the magnetic metal particle **22** contains an Fe-based alloy, an intensity of ratio a Si—K characteristic X-ray derived from the coating film **24** to a sum of an Fe—L characteristic X-ray derived from the magnetic metal particle **22** and the Si—K characteristic X-ray derived from the coating film **24** can be converted into the thickness of the coating film **24**.

[0065] A coercive force H_c of the magnetic metal particle **22** is preferably 100 A/m or less, more preferably 80 A/m or less, still more preferably 60 A/m or less, and particularly preferably 50 A/m or less. The coercive force H_c refers to a value of an external magnetic field in an opposite direction required to return a magnetized magnetic material to an unmagnetized state. That is, the coercive force H_c means a

resistance force against an external magnetic field. As the coercive force H_c of the magnetic metal particle **22** decreases, the magnetic bead **2** is less likely to aggregate even when being switched from a state in which a magnetic field is applied to a state in which no magnetic field is applied, and the magnetic bead **2** can be uniformly dispersed in a dispersion liquid. Further, even when the switching of magnetic field application is repeated, since redispersibility of the magnetic bead **2** is improved as the coercive force H_c decreases, aggregation of the magnetic bead **2** can be further limited. A lower limit value of the coercive force H_c of the magnetic metal particle **22** is not particularly limited, and is preferably 5 A/m or more from the viewpoint of ease of selection of a material suitable for a balance between performance and cost.

[0066] The coercive force H_c of the magnetic metal particle **22** can be measured by a vibrating sample magnetometer or the like in the same manner as the saturation magnetization described above. A maximum applied magnetic field when measuring the coercive force H_c is, for example, 15 kOe.

[0067] A relative permeability of the magnetic metal particle **22** is preferably 5 or more. When the relative permeability of the magnetic metal particle **22** is less than the lower limit value, the movement speed of the magnetic bead **2** decreases, and the time required for magnetic separation may increase. An upper limit value of the relative permeability of the magnetic metal particle **22** is not particularly limited, and since the magnetic bead **2** is in a powder form, the relative permeability is usually substantially 100 or less due to an influence of a demagnetizing field.

2.1. Magnetic Metal Particle

[0068] The magnetic metal particle **22** is a particle having magnetism, and preferably contains at least one of Fe, Co, and Ni as an element. In particular, from the viewpoint of obtaining a high saturation magnetization, a composition of the magnetic metal particle **22** is preferably an alloy containing Fe as a main component (an Fe-based alloy). Specifically, an Fe content is more preferably 50% or more, and still more preferably 70% or more in terms of an atomic ratio. Examples of the Fe-based alloy include an Fe—Co alloy, an Fe—Ni alloy, an Fe—Co—Ni alloy, and compounds containing Fe, Co, and Ni. From the viewpoint of obtaining a high magnetization, as the magnetic metal particle **22**, a carbonyl iron powder made of substantially 100 atomic % of Fe, an Fe—Si alloy powder, an Fe—Si—Cr alloy powder, or the like is preferably used. According to such an Fe-based alloy, it is possible to obtain the magnetic metal particle **22** having a high saturation magnetization and a high permeability even when a particle diameter is small. Accordingly, it is possible to obtain the magnetic bead **2** whose movement speed due to the action of the external magnetic field is high and whose attractive force when captured by the external magnetic field is large. As a result, the time required for magnetic separation can be shortened, and the magnetic bead **2** can be prevented from being mixed into the elution liquid and becoming a foreign substance.

[0069] The Fe-based alloy may contain one or two or more selected from the group including Cr, Nb, Cu, Al, Mn, Mo, Si, Sn, B, C, P, Ti, and Zr according to target characteristics, in addition to an element exhibiting strong magnetism alone like Fe described above. Si is a main element in an alloy powder and is also an element that promotes amorphization.

[0070] The Fe-based alloy may contain impurities within a range in which an effect of the magnetic metal particle **22** is not impaired. The impurities in the embodiment are elements that are unintentionally mixed during production of a raw material of the magnetic metal particle **22** or the magnetic bead **2**. The impurities are not particularly limited, and examples thereof include O, N, S, Na, Mg, and K.

[0071] An example of the Fe-based alloy is an alloy having a Si content of preferably 1.0 atomic % or more and 30.0 atomic % or less, more preferably 1.5 atomic % or more and 13.0 atomic % or less, and still more preferably 2.0 atomic % or more and 7.0 atomic % or less. Since such an alloy has high permeability, the saturation magnetization tends to be high.

[0072] The Fe-based alloy may contain at least one of boron (B) having a content of 5.0 atomic % or more and 16.0 atomic % or less and carbon (C) having a content of 0.5 atomic % or more and 5.0 atomic % or less. These elements are elements that promote amorphization, and contribute to stably forming an amorphous structure or a nanocrystal structure in the magnetic metal particle **22**.

[0073] Further, the Fe-based alloy preferably contains chromium (Cr) having a content of 1.0 atomic % or more and 8.0 atomic % or less. Accordingly, corrosion resistance of the magnetic metal particle **22** can be improved.

[0074] A content of impurities is preferably 1.0 atomic % or less in total. At such a level, the effect of the magnetic metal particle **22** is not impaired even when impurities are contained.

[0075] A particularly preferable example of the Fe-based alloy is an alloy containing Fe as a main component, having a Si content of 2.0 mass % or more and 9.0 mass % or less, a B content of 1.0 mass % or more and 5.0 mass % or less, a Cr content of 1.0 mass % or more and 3.0 mass % or less, and a C content of 2.0 mass % or less. Such an Fe-based alloy can have a stable amorphous structure, and thus has a low coercive force. Since the Fe content is high, the saturation magnetization is high. Further, since Cr is contained, corrosion resistance is improved, and elution of iron ions and the like can be prevented. Since iron ions may adversely influence biological material examination, it is preferable to prevent elution of iron ions.

[0076] The elements and composition of the magnetic metal particle **22** can be specified by ICP emission spectrometry defined in JIS G 1258:2014, spark emission spectrometry defined in JIS G 1253:2002, or the like. When the magnetic metal particle **22** is covered with the coating film **24**, measurement can be performed by the above-described methods after the coating film **24** is removed by a chemical or physical method. When it is difficult to remove the coating film **24**, for example, analysis can be performed, after cutting the magnetic bead **2**, on a portion of the magnetic metal particle **22** that is a core with an analysis apparatus such as an electron probe micro analyzer (EPMA) or energy dispersive X-ray spectroscopy (EDX).

[0077] A Vickers hardness of the magnetic metal particle **22** is preferably 100 or more, more preferably 300 or more, and still more preferably 800 or more. A method for measuring the hardness of the magnetic metal particle **22** is, for example, as follows. A plurality of the magnetic metal particles **22** are taken out, embedded in a resin to prepare a resin-embedded sample, and then a cross section of each magnetic metal particle **22** is exposed at a surface of the resin-embedded sample through grinding and polishing. The

cross section is subjected to indentation with a micro Vickers tester, a nanoindenter, or the like, and a hardness is measured based on a size of the indentation.

[0078] When the Vickers hardness of the magnetic metal particle **22** is less than the lower limit value, the magnetic metal particle **22** may be plastically deformed by an impact when the magnetic bead **2** undergoes collision. When plastic deformation occurs, the coating film **24** may be peeled off or detached. An upper limit value of the Vickers hardness is not particularly limited, and is preferably 3000 or less from the viewpoint of ease of selection of a material suitable for a balance of performance and cost.

[0079] A main metal structure constituting the magnetic metal particle **22** may take various forms such as a crystal structure, an amorphous structure, and a nanocrystal structure. An amorphous structure refers to a non-crystal structure in which no crystal is present, and a nanocrystal refers to a structure mainly formed of fine crystals having a crystal grain diameter of 100 nm or less. The amorphous structure and the nanocrystal structure impart a high hardness to the magnetic metal particle **22**. When the structure is the amorphous structure or the nanocrystal structure, the coercive force of the magnetic bead **2** has a particularly low value, which contributes to improvement in the redispersibility of the magnetic bead **2**. A volume fraction of the magnetic metal particle **22** having an amorphous structure or a nanocrystal structure is preferably 40% or more, and more preferably 60% or more. The volume fraction is obtained based on a result of crystal structure analysis using X-ray diffraction. Each of the crystal structure, the amorphous structure, and the nanocrystal structure may exist alone, or two or more thereof may co-exist.

[0080] The metal structure of the magnetic metal particle **22** can also be identified by crystal structure analysis using an X-ray diffraction method on the magnetic metal particle **22**. Furthermore, the metal structure can be specified by analyzing a structure observation image or a diffraction pattern obtained with a transmission electron microscope (TEM) from a cut-out sample. For example, in the case of the amorphous structure, a diffraction peak derived from a metal crystal of an α -Fe phase or the like is not observed in peak analysis of the X-ray diffraction method. In the case of the amorphous structure, a so-called halo pattern is formed in an electron diffraction pattern using TEM, and formation of a spot due to a crystal is not observed. The nanocrystal structure is formed of a crystal structure having a grain diameter of, for example, 100 nm or less, and can be checked from a TEM observation image. More accurately, an average grain diameter can be calculated by image processing or the like based on a plurality of TEM structure observation images in which there are a plurality of crystals. The crystal grain diameter can also be estimated using a Scherrer method based on a diffraction peak of a target crystal phase obtained using an X-ray diffraction method. Further, for a crystal structure having a large grain diameter, the crystal grain diameter can be measured using a method such as observing a cross section with an optical microscope or a scanning electron microscope (SEM).

[0081] In order to obtain the amorphous structure and the nanocrystal structure, it is effective to increase a cooling rate at the time of cooling after a molten raw material is pulverized when producing the magnetic metal particle **22**. Ease of forming the amorphous structure and the nanocrystal structure also depends on an alloy composition. As a specific

alloy system suitable for forming the amorphous structure and the nanocrystal structure, a composition in which one or two or more selected from the group including Cr, Si, B, C, P, Nb, and Cu are added to Fe is preferable.

[0082] The magnetic metal particle **22** is produced using a method according to a general metal powder production method. Examples of the production method include a melting process in which a metal is melted, solidified, and powdered, a chemical process in which a powder is produced by a reduction method or a carbonyl method, and a mechanical process in which a metal having a larger shape such as an ingot is mechanically pulverized to obtain a powder. Among these, the melting process is suitable for producing the magnetic metal particle **22**.

[0083] Among production methods based on the melting process, a representative production method is an atomization method. According to such a method, a molten metal having a desired composition formed by melting is sprayed to form a powder.

[0084] The atomization method causes the molten metal to collide with a fluid (liquid or gas) injected at a high speed, be rapidly quenched and solidified to form a powder, and is classified into a water atomization method, a high-pressure water atomization method, a rotary water jet atomization method, a gas atomization method, and the like depending on a difference in a type of a cooling medium and an apparatus configuration. According to the atomization method, the magnetic metal particle **22** can be efficiently produced. Further, in the high-pressure water atomization method, the rotary water jet atomization method, and the gas atomization method, a particle shape of the metal powder is closer to a spherical shape due to an action of surface tension.

2.2. Coating Film

[0085] As shown in FIG. 5, the coating film **24** covers the magnetic metal particle **22**. The coating film **24** may be formed on at least a part of a surface of the magnetic metal particle **22**, and preferably covers the entire surface.

[0086] FIG. 6 is a partially enlarged view of the coating film **24** shown in FIG. 5.

[0087] As shown in FIG. 6, the coating film **24** includes an inorganic oxide layer **242**, a base layer **244**, a gold layer **246**, and an immobilization layer **248**. That is, the coating film **24** has a multilayer structure including these layers. Each layer preferably covers an entire base thereof, but there may be a discontinuous portion.

2.2.1. Inorganic Oxide Layer

[0088] The inorganic oxide layer **242** covers the surface of the magnetic metal particle **22** and contains an inorganic oxide. Examples of the inorganic oxide include a silicon oxide, a magnesium oxide, a calcium oxide, an aluminum oxide, a titanium oxide, a zirconium oxide, a boron oxide, an yttrium oxide, and a molybdenum oxide, and a mixture of one or two or more thereof may be used. Since the inorganic oxide layer **242** containing such an inorganic oxide is porous, a large contact area with the base layer **244** can be ensured. Accordingly, adhesion of the base layer **244** to the inorganic oxide layer **242** can be improved.

[0089] Among these, the inorganic oxide is preferably a silicon oxide or a titanium oxide. Since the silicon oxide and the titanium oxide are chemically stable, oxidation and

corrosion of the magnetic metal particle **22** can be particularly reduced, and the corrosion resistance of the magnetic bead **2** can be particularly improved.

[0090] The silicon oxide is represented by a composition formula of SiO_x ($0 < x \leq 2$), and is preferably SiO_2 . The silicon oxide may form a composite oxide or a composite with one or two or more selected from the group including Al, Ti, V, Nb, Cr, Mn, Sn, and Zr.

[0091] The titanium oxide is represented by a composition formula of TiO_x ($0 < x \leq 2$), and is preferably TiO_2 . The titanium oxide may form a composite oxide or a composite with one or two or more selected from the group including Si, Al, V, Nb, Cr, Mn, Sn, and Zr.

[0092] The inorganic oxide may contain a substance (impurity) other than the inorganic oxide within a range in which an effect thereof is not impaired, for example, at a ratio of 50 mass % or less of the inorganic oxide described above. When the silicon oxide is used as the inorganic oxide, examples of the impurity include C, N, and P. A composition of the inorganic oxide can be checked by, for example, EDX analysis or Auger electron spectroscopy.

[0093] The inorganic oxide layer **242** may cover a surface of one magnetic metal particle **22**, or may cover a plurality of magnetic metal particles **22** together.

[0094] FIG. 7 is a cross-sectional view showing a modification of the magnetic bead **2** in FIG. 5.

[0095] A magnetic bead **2A** shown in FIG. 7 contains a plurality of magnetic metal particles **22**. Although not shown in FIG. 7, the inorganic oxide layer **242** covers the plurality of magnetic metal particles **22**. Although not shown in FIG. 7, the base layer **244**, the gold layer **246**, and the immobilization layer **248** are also stacked on the inorganic oxide layer **242**. In this way, the coating film **24** shown in FIG. 7 is formed. With such the magnetic bead **2A**, the same effects as those of the magnetic bead **2** shown in FIG. 5 can still be obtained. The inorganic oxide layer **242** may cover each magnetic metal particle **22**, and meanwhile, the base layer **244**, the gold layer **246**, and the immobilization layer **248** may be provided to the plurality of magnetic metal particles **22**. Alternatively, the base layer **244** may cover each magnetic metal particle **22**, and meanwhile, the gold layer **246** and the immobilization layer **248** may be provided across the plurality of magnetic metal particles **22**.

[0096] The number of the magnetic metal particles **22** in the magnetic bead **2A** is not particularly limited, and is 2 or more and 100 or less.

[0097] An average thickness of the inorganic oxide layer **242** is preferably 10 nm or more and 200 nm or less, more preferably 20 nm or more and 150 nm or less, and still more preferably 30 nm or more and 100 nm or less. Accordingly, even when the magnetic bead **2** collides with each other or collides with the inner wall of the container or the like, the inorganic oxide layer **242** can be prevented from being broken or peeled off. As a result, elution of iron ions and the like due to exposure of the magnetic metal particle **22** can be prevented. A decrease in the magnetization per unit volume of the magnetic bead **2** can be prevented, and a decrease in the movement speed of the magnetic bead **2** can be prevented.

[0098] The average thickness of the inorganic oxide layer **242** is measured in the same manner as in the above-described method for measuring the average thickness of the coating film **24**.

[0099] Examples of a method for forming the inorganic oxide layer **242** include a wet formation method such as a sol-gel method and a dry formation method such as a vapor-phase deposition method. Among these, a Stöber method, which is a type of the sol-gel method, or an atomic layer deposition (ALD) method may be preferably used. The Stöber method is a method of forming monodisperse particles by hydrolysis of a metal alkoxide. For example, when the inorganic oxide layer **242** is formed of a silicon oxide, the silicon oxide can be produced by a hydrolysis reaction of a silicon alkoxide. Before the inorganic oxide layer **242** is formed, a base thereof, for example, the surface of the magnetic metal particle **22** may be subjected to a washing treatment using water or an organic solvent.

2.2.2. Base Layer

[0100] As shown in FIG. 6, the base layer **244** is a layer that can serve as a base of the gold layer **246**. The base of the gold layer **246** refers to a surface where gold can be favorably precipitated when the gold layer **246** is formed by an electroless gold plating method. Specifically, the base layer **244** contains a reducing agent for reducing gold ions to gold or a catalyst for a reduction reaction of reducing gold ions to gold. Such the base layer **244** improves adhesion of the gold layer **246**. As a result, adhesion of the immobilization layer **248** formed at the gold layer **246** can be improved.

[0101] The reducing agent contained in the base layer **244** elutes into an electroless gold plating solution, ionizes, and releases electrons, thereby reducing gold ions to gold. Therefore, the base layer **244** containing the reducing agent enables formation of the gold layer **246** by a displacement plating method. Examples of the reducing agent include simple metals such as nickel, palladium, and copper, which exhibit a higher ionization tendency than gold, and alloys containing such metals.

[0102] When the reducing agent contained in the electroless gold plating solution causes a reduction reaction of gold ions, the catalyst contained in the base layer **244** has catalytic activity for catalyzing the reaction. Therefore, the base layer **244** containing the catalyst enables formation of the gold layer **246** by a self-catalyzed or base-catalyzed reduction plating method. Examples of the catalyst include simple metals such as nickel, palladium, and copper, and alloys containing such metals. A content of such metals in the base layer **244** is preferably more than 50 mass %, and more preferably 70 mass % or more. Examples of the reducing agent contained in the electroless gold plating solution include sulfite, thiosulfate, hydroxylamine and salts and derivatives thereof, hydrazine, amine borane compounds such as dimethylamineborane, borohydride compounds such as sodium borohydride, saccharides such as glucose, and hypophosphite salts.

[0103] As described above, the base layer **244** contains a metal element M that acts as the reducing agent or the catalyst used for reducing gold ions. Accordingly, a gold plating film can be efficiently and uniformly formed on the base layer **244**.

[0104] Examples of the metal element M include nickel, palladium, and copper, and one thereof may be used or two or more thereof may be used in combination. This can function stably as the reducing agent or the catalyst. Among these, the metal element M is preferably nickel. Nickel serves as a particularly stable reducing agent or catalyst and

contributes to efficient precipitation of gold. Therefore, since the base layer **244** contains a simple substance of nickel or a nickel alloy, the gold layer **246** can be favorably formed by an electroless gold plating method such as a displacement plating method or a self-catalyzed or base-catalyzed reduction plating method.

[0105] These reducing agents and catalysts may exist in any form. For example, the base layer **244** may be a plated film, a film formed by a vapor-phase deposition method, or a film formed by a liquid-phase deposition method. The plated film may be, for example, an electroless-plated film. Before the base layer **244** is formed, a base thereof, for example, a surface of the inorganic oxide layer **242** may be subjected to a surface treatment such as a degreasing treatment or an acid immersion treatment, or a washing treatment using water or an organic solvent.

[0106] An average thickness of the base layer **244** is not particularly limited, and is preferably 10 nm or more and 300 nm or less, more preferably 30 nm or more and 250 nm or less, and still more preferably 50 nm or more and 200 nm or less. Accordingly, it is possible to ensure a thickness for the base layer **244** to function sufficiently as a base.

[0107] The average thickness of the base layer **244** is measured in the same manner as in the above-described method for measuring the average thickness of the coating film **24**.

[0108] The average thickness of the base layer **244** may be smaller than the average thickness of the inorganic oxide layer **242**, and is preferably set to be larger than the average thickness of the inorganic oxide layer **242**. The average thickness of the base layer **244** is more preferably set to 1.2 times or more and 5.0 times or less the average thickness of the inorganic oxide layer **242**, and still more preferably set to 1.5 times or more and 4.0 times or less. Accordingly, a function of the base layer **244** as a base, that is, a function of stably forming the gold layer **246**, can be improved, and the corrosion resistance of the magnetic bead **2** can be improved. As a result, the immobilization layer **248** can be stabilized, oxidation and corrosion of the magnetic metal particle **22** can be reduced, and the magnetic bead **2** having particularly favorable nucleic acid extraction efficiency can be obtained.

[0109] The base layer **244** may have a multilayer structure. For example, on the inorganic oxide layer **242** side, a first layer may be a nickel-containing layer, and a second layer may be a palladium-containing layer. With such a multilayer structure, oxidation and corrosion of the base layer **244** can be more reliably reduced. As a result, the corrosion resistance of the magnetic bead **2** can be further improved.

2.2.3. Gold Layer

[0110] As shown in FIG. 6, the gold layer **246** is a layer that can serve as a base of the immobilization layer **248**. The base of the immobilization layer **248** refers to a surface where a compound contained in the immobilization layer **248** can be favorably bound. Specifically, examples of a material of the gold layer **246** include a simple substance of gold and an alloy containing gold. Since the gold layer **246** has high corrosion resistance, a function of reducing oxidation and corrosion of the magnetic metal particle **22** is high. Accordingly, it is possible to prevent a decrease in the magnetization of the magnetic metal particle **22** and prevent a decrease in the nucleic acid extraction efficiency due to

elution of iron ions or the like. When the magnetic bead **2** is located in a solution, a surface of the gold layer **246** is often negatively charged. Therefore, electrostatic repulsion acts between beads of the magnetic bead **2**, and aggregation of the magnetic bead **2** can be limited.

[0111] A gold content in the magnetic bead **2** is preferably 1 mass % or more and 30 mass % or less, more preferably 3 mass % or more and 25 mass % or less, and still more preferably 5 mass % or more and 20 mass % or less. Accordingly, a coverage ratio of the gold layer **246** can be sufficiently ensured, and the immobilization layer **248** can be formed evenly.

[0112] An average thickness of the gold layer **246** is preferably 0.5 nm or more and 50 nm or less, more preferably 1 nm or more and 30 nm or less, and still more preferably 2 nm or more and 20 nm or less. Accordingly, the corrosion resistance of the magnetic bead **2** can be particularly favorably ensured.

[0113] A total of average thicknesses of the inorganic oxide layer **242**, the base layer **244**, and the gold layer **246** is preferably 10 nm or more and 400 nm or less, more preferably 50 nm or more and 350 nm or less, and still more preferably 100 nm or more and 300 nm or less. Accordingly, the corrosion resistance of the magnetic bead **2** can be sufficiently ensured. In addition, the volume ratio of the coating film **24** to the entire volume of the magnetic bead **2** can be optimized, and a decrease in the magnetization per unit volume of the magnetic bead **2** can be prevented.

[0114] As described above, the method for forming the gold layer **246** may be, for example, an electroless gold plating method such as a displacement plating method or a self-catalyzed or base-catalyzed reduction plating method. Before the gold layer **246** is formed, a base thereof, for example, a surface of the base layer **244** may be subjected to a surface treatment such as a degreasing treatment or an acid immersion treatment, or a washing treatment using water or an organic solvent.

[0115] A mass ratio of a content of the metal element M to a content of gold in the magnetic bead **2** is referred to as M/Au. In the magnetic bead **2**, the mass ratio M/Au is 0.03 or more and 19.0 or less, preferably 0.10 or more and 7.00 or less, and more preferably 1.00 or more and 3.00 or less.

[0116] According to such a configuration, a quantitative balance between gold and the metal element M can be optimized. Accordingly, a balance between the thicknesses of the gold layer **246** and the base layer **244** containing the metal element M can be optimized, and thus mutual adhesion can be improved. Since coverage of the gold layer **246** can be ensured, coverage of the immobilization layer **248** supported by the gold layer **246** can be improved. Therefore, adhesion and coverage between layers can be ensured without increasing the thickness of the coating film **24**. Accordingly, it is possible to ensure a function of each layer in the magnetic bead **2** while preventing a decrease in the magnetization per unit volume of the magnetic bead **2**. As a result, elution of iron ions and the like derived from the magnetic metal particle **22**, peeling or damage of the coating film **24**, aggregation of the magnetic bead **2**, and the like can be limited. Accordingly, it is possible to obtain the magnetic bead **2** that has favorable nucleic acid extraction efficiency and can prevent mixing of a foreign substance. When the extracted nucleic acid is examined, it is possible to prevent a decrease in examination accuracy due to elution of iron ions or the like.

[0117] It is also possible to shorten the time required for magnetic separation while preventing peeling and damage of the coating film 24 when a strong external magnetic field is applied to the magnetic bead 2. In addition, an attraction force relative to the external magnetic field can be increased. Accordingly, the time required for magnetic separation can be shortened, and the magnetic bead 2 can be prevented from being mixed into the elution liquid and becoming a foreign substance.

[0118] When the mass ratio M/Au is less than the lower limit value, the content of the metal element M is low relative to the content of gold. Therefore, a film thickness and a coverage ratio of the base layer 244 serving as the base of the gold layer 246 decrease, and as a result, the coverage ratio of the gold layer 246 decreases, and a coverage ratio of the immobilization layer 248 decreases. In this case, the amount of the nucleic acid that can be extracted decreases. On the other hand, when the mass ratio M/Au exceeds the upper limit value, the content of gold is low relative to the content of the metal element M. In this case, as a result, the coverage ratio of the gold layer 246 decreases, an amount of an immobilized ligand decreases, and thus the amount of the nucleic acid that can be extracted decreases. In addition, electrostatic repulsion due to the gold layer 246 is reduced, and the dispersibility of the magnetic bead 2 decreases. In this case, nucleic acid extraction accuracy decreases due to a foreign substance incorporated into a secondary particle. Further, the corrosion resistance derived from the gold layer 246 decreases, which causes a decrease in the nucleic acid examination accuracy due to elution of gold ions or the like.

[0119] The content of gold and the content of the metal element M in the magnetic bead 2 can be specified by ICP emission spectrometry defined in JIS G 1258:2014, spark emission spectrometry defined in JIS G 1253:2002, or the like.

[0120] A mass ratio of a content of an inorganic oxide Z to the content of the metal element M in the magnetic bead 2 is referred to as Z/M. In the magnetic bead 2, the mass ratio Z/M is preferably 0.030 or more and 0.300 or less, and more preferably 0.050 or more and 0.200 or less.

[0121] According to such a configuration, a balance between the metal element M and the inorganic oxide Z can be optimized. Accordingly, a balance between the thicknesses of the base layer 244 containing the metal element M and the inorganic oxide layer 242 containing the inorganic oxide Z can be optimized, and thus mutual adhesion can be improved. In addition, coverage of the inorganic oxide layer 242 can be ensured, and adhesion between the magnetic metal particle 22 and the base layer 244 can be improved. As a result, elution of iron ions and the like derived from the magnetic metal particle 22, peeling or damage of the coating film 24, aggregation of the magnetic bead 2, and the like can be limited.

[0122] When the mass ratio Z/M is less than the lower limit value, the content of the inorganic oxide Z is low relative to the content of the metal element M. Therefore, the adhesion between the base layer 244 and the inorganic oxide layer 242 may decrease. The coverage of the inorganic oxide layer 242 may also decrease. On the other hand, when the mass ratio Z/M exceeds the upper limit value, the content of the metal element M is low relative to the content of the inorganic oxide Z. Therefore, the adhesion between the base layer 244 and the inorganic oxide layer 242 may decrease. Coverage of the base layer 244 may also decrease.

[0123] The content of the inorganic oxide Z in the magnetic bead 2 can be specified by EDX analysis, Auger electron spectroscopy, or the like on the inorganic oxide layer 242. In this case, the content of the inorganic oxide Z is calculated as a total content of a silicon oxide (SiO₂) and a titanium oxide (TiO₂). Specifically, it is assumed that Si and Ti contained in the magnetic bead 2 are all present as dioxides, and a total content of the silicon oxide (SiO₂) and the titanium oxide (TiO₂) is calculated from contents of Si and Ti. A calculation result thereof is the content of the inorganic oxide Z.

2.2.4. Immobilization Layer

[0124] The immobilization layer 248 contains a compound containing a ligand or a ligand-reactive group. The ligand is a site having a function of specifically binding to a target biological material. The ligand-reactive group is a site to which such a ligand can be bound. By including such the immobilization layer 248, the magnetic bead 2 can efficiently adsorb the target biological material.

[0125] Specific examples of the ligand include multidentate ligands such as nitrilotriacetic acid (NTA), (IDA), phenanthroline, terpyridine, iminodiacetic acid triethylenetetramine, tris(carboxymethyl)ethylenediamine, diethylenetriaminepentaacetic acid, polypyrazolylborate, bipyridine, 1,4,7-triazo-cyclononane, dimethylglyoxime, diphenylglyoxime, and derivatives thereof.

[0126] Among these, nitrilotriacetic acid (NTA) or iminodiacetic acid (IDA) is preferably used as the ligand. Such a ligand is particularly useful as the ligand.

[0127] Specific examples of the ligand-reactive group include an aldehyde group, a carboxy group, an amide group, N-hydroxysuccinimide (NHS), an amino group, a hydroxy group, $-\text{CH}=\text{CH}-$, and $-(\text{C}=\text{O})-\text{CH}=\text{CH}-$. Such a functional group is particularly useful as the ligand-reactive group.

[0128] Among these, an aldehyde group, a carboxy group, an amide group, or N-hydroxysuccinimide (NHS) is preferably used as the ligand-reactive group.

[0129] The ligand or the ligand-reactive group is bound to the gold layer 246 via a spacer provided as necessary and an Au—S bond. Since the Au—S bond is a strong bond, the Au—S bond contributes to strong immobilization of the ligand or the ligand-reactive group to the gold layer 246. Accordingly, even when various physical and chemical loads are applied to the magnetic bead 2 where the nucleic acid is adsorbed, it is possible to prevent detachment of the ligand. As a result, a decrease in efficiency of capturing the nucleic acid can be prevented, and the magnetic bead 2 that has favorable nucleic acid extraction efficiency can be obtained.

[0130] In the magnetic bead 2, the base layer 244 is provided as the base of the gold layer 246. The base layer 244 improves the adhesion of the gold layer 246. Accordingly, peeling of the gold layer 246 can be prevented, and from this viewpoint, the detachment of the ligand can also be prevented. Further, in the magnetic bead 2, the inorganic oxide layer 242 is provided as the base of the base layer 244. The inorganic oxide layer 242 improves the adhesion of the base layer 244. Accordingly, peeling of the base layer 244 can be prevented, and from this viewpoint, the detachment of the ligand can also be prevented. Therefore, the coating film 24 having the multilayer structure as described above

can prevent the detachment of the ligand and implement the magnetic bead **2** having favorable nucleic acid extraction efficiency.

[0131] The compound having a ligand or a ligand-reactive group is a compound having a thiol group or a disulfide bond that can form an Au—S bond, and is a reaction product of a reaction between a thiol derivative or a salt thereof to be described later and the gold layer **246**.

[0132] FIG. **8** is a schematic diagram showing a state in which a compound **10** having a functional site X is bound to the gold layer **246** via an Au—S bond. In FIG. **8**, a ligand or a ligand-reactive group is referred to as the “functional site X”.

[0133] The compound **10** shown in FIG. **8** has the Au—S bond, a spacer L, and the functional site X.

[0134] The spacer L is a site (linker) that connects the Au—S bond site and the functional site X. Examples of a unit structure constituting the spacer L include polyethylene glycol (PEG), polypropylene glycol (PPG), polybutylene glycol (PBG), a composite structure of a carbonate bond and PEG, a composite structure of an amide bond and PEG, methylene, and a structure in which one or two or more of these structures co-exist.

[0135] As shown in FIG. **8**, between adjacent compounds **10**, self-assembly is implemented by interaction between spacers L. Accordingly, the compound **10** is easily oriented.

As a result, it is possible to obtain the immobilization layer **248** in which the functional site X is disposed at a high density. With such the immobilization layer **248**, it is possible to increase a biological material adsorption amount per unit amount of the magnetic bead **2**. The compound **10** may have a branched structure of a molecular chain.

[0136] The following formula (1) is an example of a structure of a thiol derivative that generates the compound **10**.



[0137] [In the formula (1), x is an integer of 2 or more and 18 or less, and y is an integer of 0 or more and 100 or less. In the formula (1), R¹ represents a thiol group or a disulfide bond, and R² represents the ligand or the ligand-reactive group.]

[0138] A thiol derivative a structure having represented by the formula (1) or a salt thereof reacts with the gold layer **246** to generate the compound **10** in which the ligand or the ligand-reactive group can be disposed at a high density. Accordingly, the magnetic bead **2** having particularly favorable nucleic acid extraction efficiency can be obtained.

[0139] The structure represented by the formula (1) is all or a part of a molecular structure of the thiol derivative. Therefore, the molecular structure of the thiol derivative may include a structure represented by the formula (1) and another structure.

[0140] Examples of the salt of the thiol derivative include an alkali metal salt such as a lithium salt, a sodium salt, or

a potassium salt, and an alkaline earth metal salt such as a magnesium salt or a calcium salt.

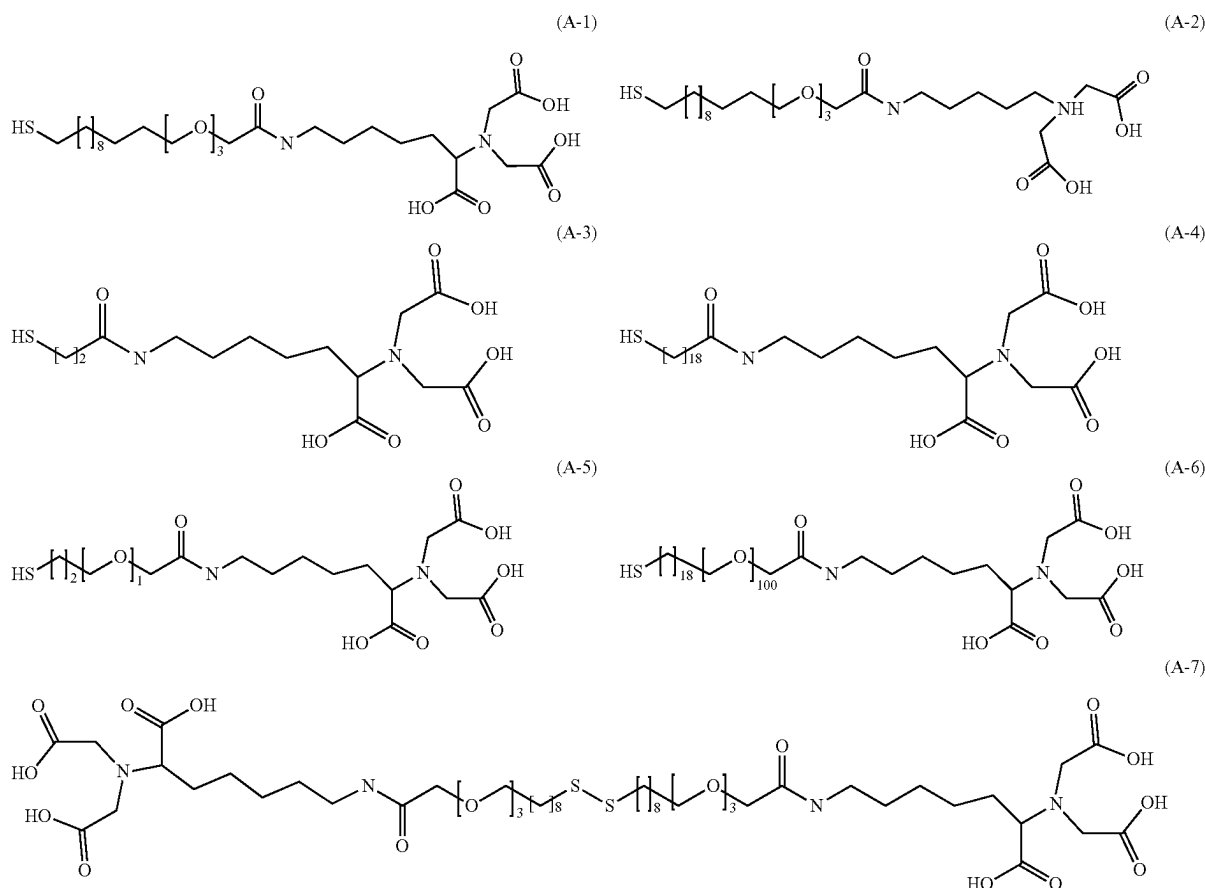
[0141] The thiol group is a binding functional group represented by —SH. The disulfide bond a bond represented by —S—S— and is converted into the thiol group by reduction. Therefore, when R¹ in the formula (1) is a disulfide bond, a thiol derivative may have two sites where R¹ is removed from the formula (1). The thiol group reacts with gold to form a strong Au—S bond. A product obtained by such a reaction is the compound **10**.

[0142] In addition, —(CH₂)_x—(C₂H₄O)_y— corresponds to the spacer L. The thiol derivative represented by the formula (1) may contain methylene or PEG as the spacer L. These spacers contribute to the self-assembly of the compound **10** when being bound to the gold layer **246**. Accordingly, the compound **10** is oriented, and the functional site X can be disposed at a high density.

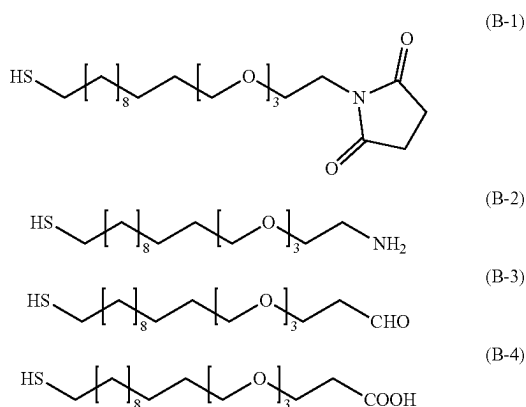
[0143] In the formula (1), x representing the number of units of methylene is 2 or more and 18 or less, preferably 3 or more and 10 or less, and more preferably 4 or more and 6 or less. Accordingly, the compound **10** is stabilized, and deterioration or the like of the immobilization layer **248** due to the magnetic separation operation and the liquid discharge operation can be prevented. When x is less than the lower limit value, a length of the spacer L is small, and thus functionality of the functional site X may decrease, that is, the ligand or the ligand-reactive group may hardly function. When the length of the spacer L is small, the self-assembly effect of the compound **10** may decrease, and the biological material adsorption amount may decrease. On the other hand, when x exceeds the upper limit value, the molecular chain of the compound **10** may be excessively long, the compound **10** may be difficult to be oriented, and the density of the functional site X may not be sufficiently high.

[0144] In the formula (1), y representing the number of units of PEG is 0 or more and 100 or less, preferably 1 or more and 30 or less, and more preferably 2 or more and 10 or less. Accordingly, the compound **10** is stabilized. When y is less than the lower limit value, the length of the spacer L is small, and thus the functionality of the functional site X may decrease. When the length of the spacer L is small, the self-assembly effect of the compound **10** may decrease. On the other hand, when y exceeds the upper limit value, the molecular chain of the compound **10** may be excessively long, the compound **10** may be difficult to be oriented, and the density of the functional site X may not be sufficiently high.

[0145] Specific examples of the thiol derivative containing the ligand include compounds represented by the following formulas (A-1) to (A-7).



[0146] Specific examples of the thiol derivative containing the ligand-reactive group include compounds represented by the following formulas (B-1) to (B-4).



[0147] Such a thiol derivative is synthesized, for example, using a method of binding linker compounds via a PEG chain or an alkyl chain using a linker compound in which various functional groups are attached to a PEG chain or an alkyl chain.

3. Effects of Embodiment

[0148] As described above, the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment includes the magnetic metal particle **22**, the inorganic oxide layer **242**, the base layer **244**, the gold layer **246**, and the immobilization layer **248**. The inorganic oxide layer **242** covers the surface of the magnetic metal particle **22** and contains the inorganic oxide Z. The base layer **244** is provided at the side opposite to the magnetic metal particle **22** of the inorganic oxide layer **242**, and contains the metal element M constituting a gold ion reducing agent or a catalyst for a gold ion reduction reaction. The gold layer **246** covers the surface of the base layer **244** and contains gold. The immobilization layer **248** contains the ligand or the ligand-reactive group that is bound to the surface of the gold layer **246** via the Au—S bond. The mass ratio M/Au of the content of the metal element M to the content of gold is 0.03 or more and 19.0 or less.

[0149] According to such configuration, it is possible to obtain the magnetic bead **2** that can prevent detachment of the ligand and reduce oxidation and corrosion of the magnetic metal particle **22** even when a load is applied. Accordingly, it is possible to obtain the magnetic bead **2** that has favorable biological material extraction efficiency and is less likely to lower examination accuracy of the extracted biological material.

[0150] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the immobilization layer **248** contains a product of reaction between a thiol derivative represented by the following formula (1) or a salt thereof and the gold layer **246**.



[0151] [In the formula (1), x is an integer of 2 or more and 18 or less, and y is an integer of 0 or more and 100 or less. In the formula (1), R¹ represents a thiol group or a disulfide bond, and R² represents the ligand or the ligand-reactive group.]

[0152] A thiol derivative having a structure represented by the formula (1) or a salt thereof can react with the gold layer **246** to generate the compound **10** in which the ligand or the ligand-reactive group can be disposed at a high density. Accordingly, the magnetic bead **2** having particularly favorable biological material extraction efficiency can be obtained.

[0153] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, R² in the formula (1) is nitrilotriacetic acid (NTA) or iminodiacetic acid (IDA), which serves as the ligand, or an aldehyde group, a carboxy group, an amide group, or N-hydroxysuccinimide (NHS), which serves as the ligand-reactive group.

[0154] Such R² is particularly useful as the ligand or the ligand-reactive group.

[0155] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the inorganic oxide Z is a silicon oxide or a titanium oxide.

[0156] Since the silicon oxide and the titanium oxide are chemically stable, oxidation and corrosion of the magnetic metal particle **22** can be particularly reduced, and the corrosion resistance of the magnetic bead **2** can be particularly improved.

[0157] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the mass ratio Z/M of the content of the inorganic oxide Z to the content of the metal element M is 0.030 or more and 0.300 or less.

[0158] According to such a configuration, the balance between the thicknesses of the base layer **244** containing the metal element M and the inorganic oxide layer **242** containing the inorganic oxide Z can be optimized, and thus mutual adhesion can be improved. In addition, the coverage of the inorganic oxide layer **242** can be ensured, and the adhesion between the magnetic metal particle **22** and the base layer **244** can be improved.

[0159] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the metal element M is nickel.

[0160] According to such a configuration, since nickel serves as a particularly stable reducing agent or catalyst and contributes to efficient precipitation of gold, the gold layer **246** can be favorably formed by an electroless gold plating method such as a displacement plating method or a self-catalyzed or base-catalyzed reduction plating method when the base layer **244** contains nickel.

[0161] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the total of the average thicknesses of the inorganic oxide layer **242**, the base layer **244**, and the gold layer **246** is 10 nm or more and 400 nm or less.

[0162] According to such a configuration, the corrosion resistance of the magnetic bead **2** can be sufficiently ensured. In addition, the volume ratio of the coating film **24** to the entire volume of the magnetic bead **2** can be optimized, and a decrease in the magnetization per unit volume of the magnetic bead **2** can be prevented.

[0163] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the average thickness of the base layer **244** is 1.2 times or more and 5.0 times or less the average thickness of the inorganic oxide layer **242**.

[0164] According to such a configuration, the function of the base layer **244** as a base, that is, the function of stably forming the gold layer **246**, can be improved, and the corrosion resistance of the magnetic bead **2** can be improved. As a result, the immobilization layer **248** can be stabilized, oxidation and corrosion of the magnetic metal particle **22** can be reduced, and the magnetic bead **2** having particularly favorable biological material extraction efficiency can be obtained.

[0165] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the average particle diameter is 0.5 μm or more and 30 μm or less.

[0166] According to such a configuration, the specific surface area of the magnetic bead **2** can be sufficiently large, and an attractive force and an adsorption force suitable for magnetic separation can be generated in the magnetic bead **2**. In addition, aggregation of the magnetic bead **2** can be prevented, and dispersibility can be improved.

[0167] In the magnetic bead **2** that is the biological material extraction magnetic bead according to the embodiment, the saturation magnetization of the magnetic metal particle **22** is 50 emu/g or more, and the coercive force of the magnetic metal particle **22** is 100 A/m or less.

[0168] According to such a configuration, since the movement speed of the magnetic bead **2** in the magnetic field can be increased, the time required for magnetic separation can be shortened. In addition, a sufficiently high adsorption force can be obtained, and therefore, when the liquid **3** is discharged in a state in which the magnetic bead **2** is fixed, the magnetic bead **2** can be prevented from being discharged together with the liquid **3**. Accordingly, it is possible to prevent a decrease in a biological material yield due to a decrease in the magnetic bead **2**. Further, even when switching of magnetic field application is repeated, aggregation of the magnetic bead **2** can be prevented.

[0169] The biological material extraction magnetic bead according to the disclosure is described above based on the shown embodiment, but the disclosure is not limited thereto. For example, the biological material extraction magnetic bead according to the disclosure may be obtained by adding any component to the embodiment. Specifically, any layer may be added between the layers in the embodiment.

EXAMPLES

[0170] Next, specific examples of the disclosure will be described.

4. Preparation of Magnetic Bead

4.1. Example 1

[0171] First, as a magnetic metal particle, a metal powder having an amorphous structure as a main structure was

prepared, which was produced using a water atomization method. The prepared metal powder was washed with ultra-pure water and ethanol.

[0172] Next, a silicon oxide (SiO_2) was formed into a film at a particle surface of the magnetic metal particle using a Stöber method to obtain an inorganic oxide layer. As a silicon alkoxide, tetraethoxysilane (TEOS) was used. Subsequently, the magnetic metal particle where the inorganic oxide layer was formed was subjected to a degreasing treatment and an acid immersion treatment.

[0173] Next, nickel was formed into a film at a surface of the inorganic oxide layer using an electroless nickel plating method to obtain a base layer. The magnetic metal particle where the base layer was formed was subjected to a degreasing treatment.

[0174] Next, gold was formed into a film at a surface of the base layer using a displacement type electroless gold plating method to obtain a gold layer. The magnetic metal particle where the gold layer was formed was washed with ultra-pure water and ethanol.

[0175] Next, an ethanol solution was prepared by dissolving a thiol derivative having a structure represented by a symbol shown in Table 1 in ethanol. Subsequently, the magnetic metal particle where the gold layer was formed was immersed in the ethanol solution and stirred. Accordingly, the thiol derivative was reacted with the gold layer to obtain an immobilization layer.

[0176] Configurations of a magnetic bead (an average particle diameter of the magnetic bead, a saturation magnetization and a coercive force of the magnetic metal particle, a formation condition of a coating film, and the like) are as shown in Table 1. The symbol representing the thiol derivative shown in Table 1 corresponds to the formula (A-1) described above.

4.2. Examples 2 to 8 and Comparative Examples 1 and 2

[0177] Magnetic beads were obtained in the same manner as in Example 1 except that configurations of the magnetic beads were as shown in Table 1.

4.3. Examples 9 to 15 and Comparative Examples 3 and 4

[0178] Magnetic beads were obtained in the same manner as in Example 1 except that configurations of the magnetic beads were as shown in Table 2.

4.4. Examples 16 to 19

[0179] Magnetic beads were obtained in the same manner as in Example 1 except that configurations of the magnetic beads were as shown in Table 3.

4.5. Comparative Example 5

[0180] A magnetic bead was obtained in the same manner as in Example 1 except that the inorganic oxide layer was omitted and configurations of the magnetic bead were as shown in Table 3.

4.6. Comparative Example 6

[0181] A magnetic bead was obtained in the same manner as in Example 1 except that the base layer was omitted and configurations of the magnetic bead were as shown in Table 3.

4.7. Comparative Example 7

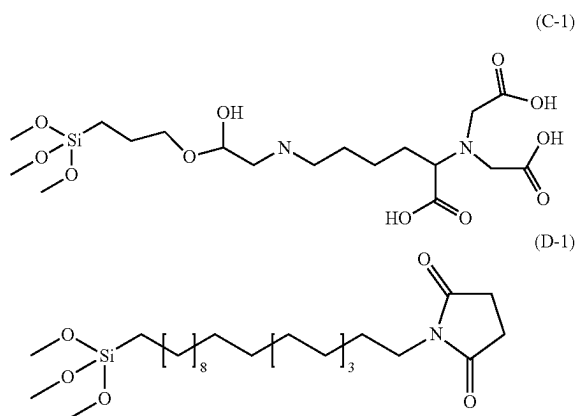
[0182] A magnetic bead was obtained in the same manner as in Example 1 except that a gold colloid particle was used instead of the magnetic metal particle. The gold colloid particle is a particle containing only gold and does not include the inorganic oxide layer and the base layer.

4.8. Comparative Example 8

[0183] A magnetic bead was obtained in the same manner as in Example 1 except that a ferrite-containing particle was used instead of the magnetic metal particle, and configurations of the magnetic bead were as shown in Table 3. The ferrite-containing particle is a particle obtained by dispersing a ferrite particle in silica.

4.9. Comparative Examples 9 and 10

[0184] Magnetic beads were obtained in the same manner as in Example 1 except that silane coupling agents were used instead of the thiol derivative, and configurations of the magnetic beads as shown in Table 3. Symbols representing silane coupling agents shown in Table 3 correspond to silane coupling agents represented by the following formulas (C-1) and (D-1).



4.6. Examples 20 to 29

[0185] Magnetic beads were obtained in the same manner as in Example 1 except that configurations of the magnetic beads were as shown in Table 4. Symbols representing thiol derivatives shown in Table 4 correspond to thiol derivatives represented by the formulas (A-2) to (A-7) and the formulas (B-1) to (B-4) described above.

5. Evaluation of Magnetic Bead

5.1. Purification Yield of Biological Material

[0186] First, a protein A as a target biological material was dispersed in pure water to prepare a specimen aqueous solution in a microtube. A concentration of the protein A was 50 $\mu\text{g/mL}$.

[0187] Next, the magnetic bead of each of Examples and Comparative Examples was dispersed in the specimen aqueous solution such that a concentration was 30 mass %.

[0188] Next, a magnet was brought close to a side surface of the microtube, a magnetic field was applied, and then a supernatant was removed. Then, the magnetic bead remaining in the microtube was collected. Accordingly, the target biological material was adsorbed to the magnetic bead. Subsequently, the collected magnetic bead was dispersed in a PBS buffer aqueous solution to prepare a dispersion liquid. The PBS buffer aqueous solution was a phosphate buffer containing NaCl having a concentration of 137 mmol/L, Na_2HPO_4 having a concentration of 8.1 mmol/L, KCl having a concentration of 2.7 mmol/L, and KH_2PO_4 having a concentration of 1.5 mmol/L.

[0189] Next, the prepared dispersion liquid was stirred by inversion stirring for 30 minutes, followed by collection of a supernatant and addition of a BCA assay testing solution. Thereafter, absorbance of the dispersion liquid to which the testing solution was added was measured with a UV-vis measuring instrument (U-3900H type spectrophotometer). Then, using a value of abs. -562 nm of the specimen aqueous solution alone as a reference value, a ratio of a value of abs. -562 nm of the supernatant obtained using the magnetic bead of each of Examples and Comparative Examples was calculated. As the ratio is closer to 1.0, a purification yield of the biological material by the magnetic bead is larger. Subsequently, the calculated ratio was evaluated in view of the following evaluation criteria. Evaluation results are shown in Tables 1 to 4.

[0190] A: the ratio of the value of abs. -562 nm of the supernatant to the reference value is 0.8 or more

[0191] B: the ratio of the value of abs. -562 nm of the supernatant to the reference value is 0.7 or more and less than 0.8

[0192] C: the ratio of the value of abs. -562 nm of the supernatant to the reference value is 0.6 or more and less than 0.7

[0193] D: the ratio of the value of abs. -562 nm of the supernatant to the reference value is 0.3 or more and less than 0.6

[0194] E: the ratio of the value of abs. -562 nm of the supernatant to the reference value is less than 0.3

5.2. Redispersibility

[0195] First, a dispersion liquid was prepared by dispersing the magnetic bead of each of Examples and Comparative Examples in a PBS buffer aqueous solution to have a concentration of 30 mass %. The PBS buffer aqueous solution was the same as that used in 5.1.

[0196] Next, the prepared dispersion liquid was set in a laser diffraction and scattering particle diameter distribution measuring apparatus, and the average particle diameter D50 on a volume basis was measured. A measured value of the obtained average particle diameter D50 is taken as a value before separation of the target biological material, and is taken as a reference value.

[0197] Next, the magnetic bead was dispersed as in 5.1. in a specimen aqueous solution prepared in the same manner as in 5.1., and magnetic separation was performed to collect the magnetic bead where the biological material is adsorbed. Thereafter, in the same manner as in 5.1., the collected magnetic bead was dispersed in a PBS buffer aqueous solution to prepare a dispersion liquid. Then, the average

particle diameter D50 on a volume basis was measured in the same manner as described above, and a measured value thereof was set as a value after separation of the target biological material.

[0198] Next, a ratio of the value after separation to the reference value was calculated. As the ratio is closer to 1.0, redispersibility of the magnetic bead is more favorable. Subsequently, the calculated ratio was evaluated in view of the following evaluation criteria. Evaluation results are shown in Tables 1 to 4.

[0199] A: the ratio of the value after separation to the reference value is less than 1.1

[0200] B: the ratio of the value after separation to the reference value is 1.1 or more and less than 1.3

[0201] C: the ratio of the value after separation to the reference value is 1.3 or more and less than 1.5

[0202] D: the ratio of the value after separation to the reference value is 1.5 or more and less than 2.0

[0203] E: the ratio of the value after separation to the reference value is 2.0 or more

5.3. Corrosion Resistance

[0204] First, a dispersion liquid was prepared by dispersing the magnetic bead of each of Examples and Comparative Examples in a PBS buffer aqueous solution to have a concentration of 30 mass %. The PBS buffer aqueous solution was the same as that used in 5.1.

[0205] Next, the prepared dispersion liquid was allowed to stand for 10 days under conditions of an atmospheric pressure of 1013 hPa and a temperature of 70° C., and then an element concentration of a supernatant was measured using an ICP analyzer 5100 ICP-OES manufactured by Agilent Technologies, Ltd.

[0206] Then, a difference from the element concentration of the supernatant was calculated using a value of an element concentration of the PBS buffer aqueous solution alone as a reference value. As a difference between an ion concentration of the supernatant and an ion concentration of the reference value is smaller, a magnetic component elution amount is smaller. Evaluation results are shown in Tables 1 to 4.

[0207] A: the difference in the ion concentration is less than 10 ppm

[0208] B: the difference in the ion concentration is 10 ppm or more and less than 20 ppm

[0209] C: the difference in the ion concentration is 20 ppm or more and less than 35 ppm

[0210] D: the difference in the ion concentration is 35 ppm or more and less than 100 ppm

[0211] E: the difference in the ion concentration is 100 ppm or more

5.4. Magnetic Separation Rate

[0212] First, a dispersion liquid was prepared by dispersing the magnetic beads of each of Examples and Comparative Examples in pure water at 25° C. to have a concentration of 0.1 mass %. Next, the dispersion liquid was charged in a spectroscopic cell and stirred by ultrasonic irradiation or stirred by a vortex mixer. A stirring time was 1 minute. Next, the spectroscopic cell subjected to the stirring treatment was quickly set in a cell holder of a spectrophotometer. A magnet was attached to the cell holder in advance according to a position where the spectroscopic cell was disposed. A short-

est distance between the magnet and an outer wall of the spectroscopic cell set in the cell holder was 2.0 mm, and a magnet having a surface magnetic flux density of 180 mT was used as the magnet.

[0213] Next, simultaneously with the start of standing of the spectroscopic cell, measurement of absorbance at a wavelength of 550 nm in the spectroscopic cell was started. Then, a time until the measured absorbance was attenuated to 10% of initial absorbance was measured, and a measurement result thereof was used as an evaluation index for evaluating a magnetic separation rate. The obtained evaluation index was evaluated in view of the following evaluation criteria. Evaluation results are shown in Tables 1 to 4.

[0214] A: the evaluation index of the magnetic separation rate is shorter than 20 seconds

[0215] B: the evaluation index of the magnetic separation rate is 20 seconds or longer and shorter than 30 seconds

[0216] C: the evaluation index of the magnetic separation rate is 30 seconds or longer and shorter than 45 seconds

[0217] D: the evaluation index of the magnetic separation rate is 45 seconds or longer and shorter than 60 seconds

[0218] E: the evaluation index of the magnetic separation rate is 60 seconds or longer

TABLE 1

				Example 1	Example 2	Example 3	Example 4	Example 5	
Magnetic bead		Average particle diameter	μm	3.2	3.2	3.2	3.2	3.2	
Magnetic metal particle		Saturation magnetization	emu/g	108	108	108	108	108	
		Coercive force	A/m	36	36	36	36	36	
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	1.5	1.5	1.4	1.4	1.1
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	17.7	17.2	16.6	16.1	12.8
		Cu	Content	mass %	—	—	—	—	—
	Gold layer	Au	Content	mass %	1.1	3.6	6.9	10.1	28.0
			Average thickness	nm	3	10	20	30	100
	Total of average thicknesses of inorganic oxide layer, base layer, and gold layer			nm	130	130	120	120	200
	Thickness ratio of inorganic oxide layer to base layer			—	3.3	4.0	3.5	3.3	3.3
X/M			—	0.085	0.087	0.084	0.087	0.086	
M/Au			—	16.09	4.78	2.41	1.59	0.46	
Immobilization layer	Thiol derivative containing ligand	Thiol derivative containing ligand- reactive group	—	A-1	A-1	A-1	A-1	A-1	
		Silane coupling agent containing ligand	—	—	—	—	—	—	
		Silane coupling agent containing ligand-reactive group	—	—	—	—	—	—	
		Content of thiol derivative	mass %	1.0	1.0	0.9	0.9	0.7	
		Evaluation result of magnetic bead	Purification yield	—	C	B	A	A	B
			Redispersibility	—	C	B	B	A	A
			Corrosion resistance	—	C	C	B	A	A
			Magnetic separation rate	—	A	A	A	A	A
				Example 6	Example 7	Example 8	Compar- ative Exam- ple 1	Compar- ative Exam- ple 2	
Magnetic bead		Average particle diameter	μm	3.2	3.2	2.5	3.2	3.2	
Magnetic metal particle		Saturation magnetization	emu/g	108	108	84	108	108	
		Coercive force	A/m	36	36	88	36	36	

TABLE 1-continued

Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	0.7	0.2	1.7	1.5	0.3
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	8.6	3.6	—	17.7	1.8
		Cu	Content	mass %	—	—	15.0	—	—
	Gold layer	Au	Content	mass %	51.6	79.9	8.0	0.9	80.9
			Average thickness	nm	250	750	25	3	750
	Total of average thicknesses of inorganic oxide layer, base layer, and gold layer			nm	350	800	120	130	800
	Thickness ratio of inorganic oxide layer to base layer			—	2.4	2.0	3.0	3.3	3.3
	X/M			—	0.081	0.056	0.113	0.085	0.167
	M/Au			—	0.17	0.05	1.88	19.67	0.02
Immobilization layer	Thiol derivative containing ligand- reactive group	Silane coupling agent containing ligand	—	A-1	A-1	A-1	A-1	A-1	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
		Silane coupling agent containing ligand- reactive group	—	—	—	—	—	—	
	Content of thiol derivative	mass %	0.5	0.2	1.5	1.0	0.2		
Evaluation result of magnetic bead	Purification yield			—	B	C	A	D	D
	Redispersibility			—	B	A	A	D	A
	Corrosion resistance			—	A	A	A	D	A
	Magnetic separation rate			—	A	A	A	A	A

TABLE 2

				Example 9	Example 10	Example 11	Example 12	Example 13
Magnetic bead		Average particle diameter	μm	3.2	3.2	3.2	3.2	3.2
Magnetic metal particle		Saturation magnetization	emu/g	108	108	108	108	108
		Coercive force	A/m	36	36	36	36	36
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	—	—	—	—
		TiO ₂	Content	mass %	2.7	2.6	2.5	1.4
	Base layer	Ni	Content	mass %	17.0	16.4	15.9	12.7
		Cu	Content	mass %	—	—	—	—
	Gold layer	Au	Content	mass %	3.5	6.8	10.0	27.8
			Average thickness	nm	10	20	30	100
	Total of average thicknesses of inorganic oxide layer, base layer, and gold layer			nm	220	210	200	150
	Thickness ratio of inorganic oxide layer to base layer			—	4.5	4.2	3.2	3.5
	X/M			—	0.159	0.159	0.157	0.157
	M/Au			—	4.86	2.41	1.59	0.46
Immobilization layer	Thiol derivative containing ligand- reactive group	Silane coupling agent containing ligand	Silane coupling agent containing ligand- reactive group	Content of thiol derivative	—	A-1	A-1	A-1
					—	—	—	—
					—	—	—	—
					—	—	—	—
					—	—	—	—

TABLE 2-continued

			Silane coupling agent containing ligand-reactive group	—	—	—	—	—	—
			Content of thiol derivative	mass %	1.0	0.9	0.9	0.7	0.5
Evaluation result of magnetic bead	Purification yield			—	B	A	A	B	B
	Redispersibility			—	B	B	A	A	B
	Corrosion resistance			—	C	B	A	A	A
	Magnetic separation rate			—	A	A	A	A	A
					Example 14	Example 15	Comparative Example 3	Comparative Example 4	
	Magnetic bead		Average particle diameter	μm	3.2	4.5	3.2	3.2	
	Magnetic metal particle		Saturation magnetization	emu/g	108	120	108	108	
			Coercive force	A/m	36	28	36	36	
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	—	—	—	—	
		TiO ₂	Content	mass %	0.6	2.4	1.5	0.3	
	Base layer	Ni	Content	mass %	3.4	—	17.7	1.8	
		Cu	Content	mass %	—	14.0	—	—	
	Gold layer	Au	Content	mass %	79.2	12.0	0.9	80.9	
			Average thickness	nm	750	35	3	750	
	Total of average thicknesses of inorganic oxide layer, base layer, and gold layer			nm	800	180	200	800	
	Thickness ratio of inorganic oxide layer to base layer			—	2.5	3.0	3.2	2.6	
	X/M			—	0.176	0.171	0.085	0.167	
	M/Au			—	0.04	1.17	19.67	0.02	
Immobilization layer	Thiol derivative containing ligand	Thiol derivative containing ligand-reactive group	—	A-1	A-1	A-1	A-1		
		Thiol derivative containing ligand-reactive group	—	—	—	—	—		
		Silane coupling agent containing ligand	—	—	—	—	—		
		Silane coupling agent containing ligand-reactive group	—	—	—	—	—		
		Content of thiol derivative	mass %	0.2	1.5	1.0	0.2		
				—	C	A	D	D	
Evaluation result of magnetic bead	Purification yield			—	A	A	D	A	
	Corrosion resistance			—	A	A	D	A	
	Magnetic separation rate			—	A	A	A	A	

TABLE 3

				Example 16	Example 17	Example 18	Example 19	Comparative Example 5
Magnetic bead		Average particle diameter	μm	3.2	3.2	3.2	3.2	3.2
Magnetic metal particle		Saturation magnetization	emu/g	108	108	108	108	108
		Coercive force	A/m	36	36	36	36	36

TABLE 3-continued

Ferrite-containing particle			Average particle diameter	μm	—	—	—	—	—
			Saturation magnetization	emu/g	—	—	—	—	—
Gold colloid particle			Average particle diameter	μm	—	—	—	—	—
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	2.5	0.7	0.3	2.4	—
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	19.8	8.6	11.9	6.5	12.5
		Cu	Content	mass %	—	—	—	—	—
	Gold layer	Au	Content	mass %	10.1	12.5	8.9	10.1	16.0
Total of average thicknesses of inorganic oxide layer, base layer, and gold layer				nm	250	100	130	80	100
X/M				—	0.126	0.081	0.025	0.369	0.000
M/Au				—	1.96	0.69	1.34	0.64	0.78
Immobilization layer	Thiol derivative containing ligand			—	A-1	A-1	A-1	A-1	A-1
		Thiol derivative containing ligand-reactive group		—	—	—	—	—	—
		Silane coupling agent containing ligand		—	—	—	—	—	—
		Silane coupling agent containing ligand-reactive group		—	—	—	—	—	—
		Content of thiol derivative		mass %	0.9	1.0	0.9	0.9	0.2
Evaluation result of magnetic bead	Purification yield			—	A	B	C	C	E
	Redispersibility			—	A	A	C	C	E
	Corrosion resistance			—	A	A	C	C	E
	Magnetic separation rate			—	A	A	A	A	B
					Comparative Example 6	Comparative Example 7	Comparative Example 8	Comparative Example 9	Comparative Example 10
Magnetic bead			Average particle diameter	μm	3.2	—	—	3.2	3.2
Magnetic metal particle			Saturation magnetization	emu/g	108	—	—	108	108
			Coercive force	A/m	36	—	—	36	36
Ferrite-containing particle			Average particle diameter	μm	—	—	2.8	—	—
			Saturation magnetization	emu/g	—	—	11	—	—
Gold colloid particle			Average particle diameter	μm	—	0.1	—	—	—

TABLE 3-continued

Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	2.1	—	1.5	1.4	1.4
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	—	—	16.0	16.6	16.6
		Cu	Content	mass %	—	—	—	—	—
	Gold layer	Au	Content	mass %	79.9	80.9	79.9	10.1	10.1
Total of average thicknesses of inorganic oxide layer, base layer, and gold layer				nm	30	0	130	120	120
X/M				—	—	—	0.094	0.084	0.084
M/Au				—	0.00	0.00	0.20	1.64	1.64
Immobilization layer	Thiol derivative containing ligand	Thiol derivative containing ligand-reactive group	Silane coupling agent containing ligand	—	A-1	A-1	A-1	—	—
				—	—	—	—	—	—
				—	—	—	—	C-1	—
				—	—	—	—	—	D-1
				—	—	—	—	—	—
				—	—	—	—	—	—
Content of thiol derivative				mass %	1.5	0.2	1.5	1.0	0.2
Evaluation result of magnetic bead	Purification yield			—	E	C	D	E	E
	Redispersibility			—	E	E	A	C	B
	Corrosion resistance			—	A	A	C	A	A
	Magnetic separation rate			—	B	E	D	A	A

TABLE 4

					Example 20	Example 21	Example 22	Example 23	Example 24
Magnetic bead			Average particle diameter	μm	8.5	8.5	8.5	8.5	8.5
Magnetic metal particle			Saturation magnetization	emu/g	200	200	200	200	200
			Coercive force	A/m	25	25	25	25	25
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	1.4	1.4	1.4	1.4	1.4
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	16.1	16.1	16.1	16.1	16.1
		Cu	Content	mass %	—	—	—	—	—
	Gold layer	Au	Content	mass %	10.1	10.1	10.1	10.1	10.1
		Total of average thicknesses of inorganic oxide layer, base layer, and gold layer				nm	130	130	130
			X/M	—	0.087	0.087	0.087	0.087	0.087
			M/Au	—	1.59	1.59	1.59	1.59	1.59
	Immobilization layer	Thiol derivative containing ligand	—	A-2	A-3	A-4	A-5	A-6	
Thiol derivative containing ligand- reactive group			—	—	—	—	—	—	
Silane coupling agent containing ligand			—	—	—	—	—	—	
Silane coupling agent containing ligand-reactive group			—	—	—	—	—	—	
Content of thiol derivative			mass %	0.9	0.9	0.9	0.9	0.9	
Purification yield			—	A	A	A	A	A	
Redispersibility			—	A	A	A	A	A	
Evaluation result of magnetic bead	Corrosion resistance	—	A	A	A	A	A		
	Magnetic separation rate	—	A	A	A	A	A		

TABLE 4-continued

					Example 25	Example 26	Example 27	Example 28	Example 29
Magnetic bead			Average particle diameter	μm	8.5	8.5	8.5	8.5	8.5
Magnetic metal particle			Saturation magnetization	emu/g	200	200	200	200	200
			Coercive force	A/m	25	25	25	25	25
Coating film formation condition	Inorganic oxide layer	SiO ₂	Content	mass %	1.4	1.4	1.4	1.4	1.4
		TiO ₂	Content	mass %	—	—	—	—	—
	Base layer	Ni	Content	mass %	16.1	16.1	16.1	16.1	16.1
		Cu	Content	mass %	—	—	—	—	—
	Gold layer	Au	Content	mass %	10.1	10.1	10.1	10.1	10.1
	Total of average thicknesses of inorganic oxide layer, base layer, and gold layer			nm	130	130	130	130	130
	X/M			—	0.087	0.087	0.087	0.087	0.087
	M/Au			—	1.59	1.59	1.59	1.59	1.59
	Immobilization layer	Thiol derivative containing ligand			—	A-7	—	—	—
Thiol derivative containing ligand-reactive group			—	—	B-1	B-2	B-3	B-4	
Silane coupling agent containing ligand			—	—	—	—	—	—	
Silane coupling agent containing ligand-reactive group			—	—	—	—	—	—	
Content of thiol derivative			mass %	0.9	0.9	0.9	0.9	0.9	
Evaluation result of magnetic bead	Purification yield			—	A	A	A	A	A
	Redispersibility			—	A	A	A	A	A
	Corrosion resistance			—	A	A	A	A	A
	Magnetic separation rate			—	A	A	A	A	A

[0219] From the evaluation results shown in Tables 1 to 4, the following can be recognized.

[0220] In the case of the magnetic bead of each Example, the purification yield of the biological material was large and the redispersibility was also favorable even after the magnetic separation operation

[0221] the magnetic bead of each Example had favorable corrosion resistance even in a state of being immersed in the PBS buffer aqueous solution containing an acid for 10 days

[0222] it was also recognized that the magnetic bead of each Example had a sufficiently high magnetic separation rate.

[0223] From the above results, it is recognized that, according to the present disclosure, it is possible to provide a biological material extraction magnetic bead that has favorable biological material extraction efficiency and is less likely to lower examination accuracy of the extracted biological material.

What is claimed is:

1. A biological material extraction magnetic bead comprising:

- a magnetic metal particle;
- an inorganic oxide layer that covers a surface of the magnetic metal particle and contains an inorganic oxide;
- a base layer that is provided on a side opposite to the magnetic metal particle of the inorganic oxide layer and

contains a metal element forming a gold ion reducing agent or a catalyst for a gold ion reduction reaction;

a gold layer that covers a surface of the base layer and contains gold; and

an immobilization layer that is bound to the gold layer via an Au—S bond and contains a ligand or a ligand-reactive group, wherein

a mass ratio M/Au of a content of the metal element to a content of the gold is 0.03 or more and 19.0 or less.

2. The biological material extraction magnetic bead according to claim 1, wherein

the immobilization layer contains a product of a reaction between a thiol derivative represented by the following formula (1) or a salt thereof and the gold layer:



in the formula (1), x is an integer of 2 or more and 18 or less, and y is an integer of 0 or more and 100 or less, and in the formula (1), R¹ represents a thiol group or a disulfide bond, and R² represents the ligand or the ligand-reactive group.

3. The biological material extraction magnetic bead according to claim 2, wherein

R² in the formula (1) is nitrilotriacetic acid (NTA) or iminodiacetic acid (IDA), which serves as the ligand, or an aldehyde group, a carboxy group, an amide group, or N-hydroxysuccinimide (NHS), which serves as the ligand-reactive group.

4. The biological material extraction magnetic bead according to claim 1, wherein

the inorganic oxide is a silicon oxide or a titanium oxide.

5. The biological material extraction magnetic bead according to claim 4, wherein

a mass ratio Z/M of a content of the inorganic oxide to the content of the metal element is 0.030 or more and 0.300 or less.

6. The biological material extraction magnetic bead according to claim 1, wherein

the metal element is nickel.

7. The biological material extraction magnetic bead according to claim 1, wherein

a total of average thicknesses of the inorganic oxide layer, the base layer, and the gold layer is 10 nm or more and 400 nm or less.

8. The biological material extraction magnetic bead according to claim 1, wherein

an average thickness of the base layer is 1.2 times or more and 5.0 times or less an average thickness of the inorganic oxide layer.

9. The biological material extraction magnetic bead according to claim 1, wherein

an average particle diameter is 0.5 μm or more and 30 μm or less.

10. The biological material extraction magnetic bead according to claim 1, wherein

a saturation magnetization of the magnetic metal particle is 50 emu/g or more, and

a coercive force of the magnetic metal particle is 100 A/m or less.

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